Menna Mohammed Abdelbaky

Data Scientist Student

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EDUCATION

Alexandria University Alexandria,Egypt

Bachelor of Computing and Data Science (CGPA: 3.97 / 4.00) SEP 2023 – July 2027

Faculty of computer and Data Science

PROJECTS

CKD Multi-Stage Classification | Python • Scikit-learn • PCA • GridSearchCV • Pandas • NumPy

- Developed an end-to-end machine learning pipeline for five-stage CKD classification using SVM, PCA, and feature selection techniques.
- Optimized model performance with GridSearchCV and 5-fold cross-validation, achieving **96.08% accuracy** and **96.69% macro recall**.
- Applied data preprocessing, feature engineering, dimensionality reduction, and comprehensive model evaluation.

Crime Analysis & Prediction | Python • Logistic Regression • K-Medoids • Plotly

- Analyzed historical crime data through data cleaning, EDA, and geospatial visualization to uncover crime patterns.
- Developed a multi-class Logistic Regression model and evaluated its performance using multiple classification metrics.
- Applied K-Medoids clustering with Gower distance to identify representative crime profiles and hidden patterns.

Alexandria Weather Data Analysis | Python • Streamlit • MongoDB Atlas • Plotly • BeautifulSoup

- Built an end-to-end data science project involving web scraping, preprocessing, exploratory data analysis, and temperature prediction.
- Developed a Multiple Linear Regression model (**R^2 = 0.67**) and stored processed data in MongoDB Atlas.
- Created an interactive Streamlit dashboard to visualize historical weather trends and insights.

Egypt Weather Analysis using MapReduce | Python • MRJob • Pandas

- Processed large-scale historical weather data using Python and the MapReduce programming model.
- Performed feature engineering, data preprocessing, and exploratory data analysis to identify long-term climate trends.
- Implemented distributed processing with MRJob to compute yearly maximum temperatures efficiently.

EXPERIENCE

Information Technology Institute (ITI) Alexandria

Applied AI Development August 2026

Faculty of Computer & Data Science Alexandria

IOT Summer Internship July 2025 – August 2025

- Participated in hands-on IoT training covering sensors, embedded systems, and IoT applications.
- Collaborated with team members to build and test IoT-based projects.
- Gained practical experience in hardware integration and IoT development concepts.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C++

Machine Learning: Scikit-learn, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning, PCA

Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Probability & Statistics

Visualization: Power BI, Matplotlib, Plotly

Databases: SQL Server, MongoDB, Supabase

Developer Tools: Git, GitHub, Linux, VS Code, AWS